NetOps Terminal 1.1.0 User Guide

A practical guide for Windows users who connect to network devices, organize a device list, and work in terminal sessions.

> **About the pictures:** These are labeled illustrations of the version 1.1.0 controls, not screenshots of a real device or network. Example names and addresses are fictional. The installed app may differ slightly with window size.

1. Install and start

NetOps Terminal 1.1.0 runs on Windows 10/11 x64. The installer includes the .NET runtime.

1. Open the v1.1.0 release. If the repository is private, sign in to an account with access.
2. Download **NetOpsTerminal-1.1.0-win-x64.msi** and its **.sha256** file from **Assets**. Verify the downloaded file if your organization requires a checksum check.
3. Close any running NetOps Terminal window, then run the MSI. Approve the Windows administrator prompt.
4. Read and accept the software agreement. The default installation folder is **C:\Program Files\NetOps Terminal**; use **Browse** if you need another location.
5. Keep or clear the separate **Create a desktop shortcut** and **Add NetOps Terminal to the Start menu** checkboxes, then finish the wizard.
6. Open **NetOps Terminal** from Start or your Desktop shortcut.

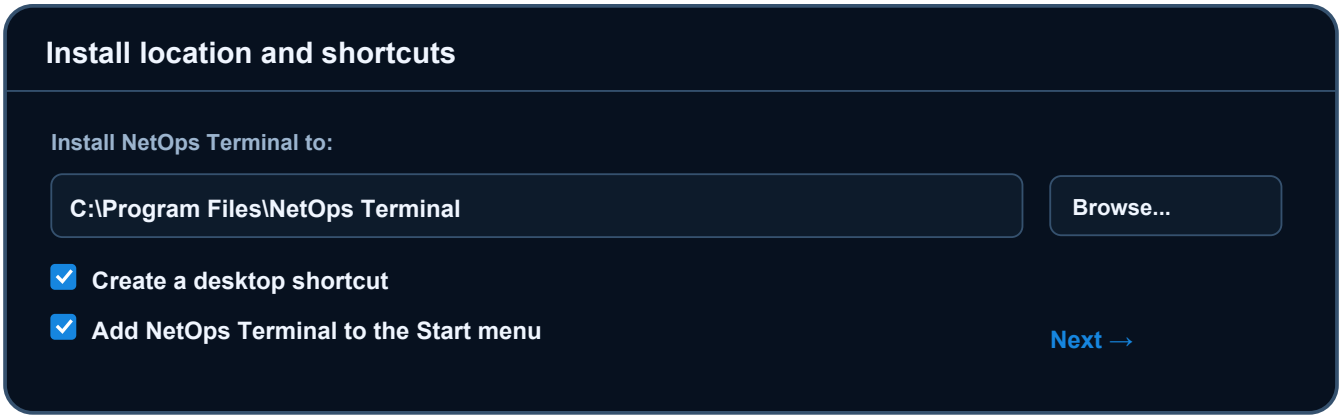


Illustration of the installer shortcut choices

The installer is unsigned, so Windows may display a publisher warning. Obtain the package from the intended release and verify its checksum before proceeding. The optional portable ZIP is on the same release page. There is no automatic updater.

Upgrading from 1.0.0: Back up **%LOCALAPPDATA%\NetOpsTerminal** for each Windows account before launching 1.1.0. The device database upgrade is forward only: 1.0.0 cannot read the upgraded database. Saved passwords are held separately in Windows Credential Manager. See the release guide.

2. Find your way around

The left rail opens Device Explorer, Quick Connect, Add device, local shells, Credentials, Import, Export, and Settings. Hover over a rail icon to see its name. The **?** button in the title bar opens Help & Shortcuts. Device Explorer contains a search box, **All**, **Favorites**, and **Recent** filters, sites, organization folders, and device rows.

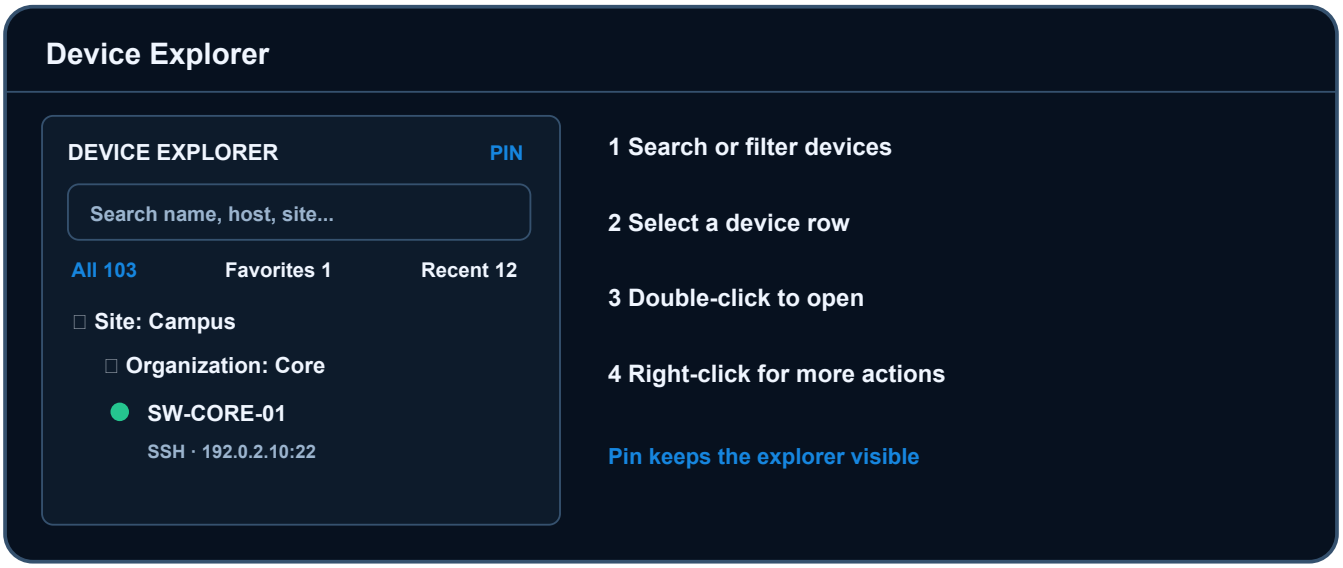


Illustration of Device Explorer and its main controls

The small text below a device name shows its protocol and endpoint. **WEB** means web only; **CLI+WEB** means both access methods. Environment badges use short labels such as **PROD**, **TEST**, **LAB**, and **DEV**. The pin keeps Device Explorer open while you work. When unpinned, the explorer may close after a successful connection or opening a local shell; sites and folders retain the expanded or collapsed state you chose.

Select a device to use the **Connect** or **Open web interface** action shown for that device. Double-click uses the saved default action. Right-click a device for its other available actions.

3. Connect once with Quick Connect

Use **Quick Connect** on the left rail, or press **Ctrl+N**, when you need a session without saving a device.

1. Choose **SSH**, **Telnet**, **Serial**, or **Raw TCP**.
2. Fill in the fields shown for that protocol. SSH uses port **22** and Telnet uses port **23**; these fields are fixed in version 1.1.0. Raw TCP uses a configurable port. Serial uses a local COM port and serial settings.
3. For SSH, enter the hostname or IP, username, password, and timeout; click **Connect**. The Quick Connect password is kept only for that terminal tab.
4. For Telnet, read and select **I understand this Telnet session is plaintext** before connecting. Telnet sends credentials and output without encryption.

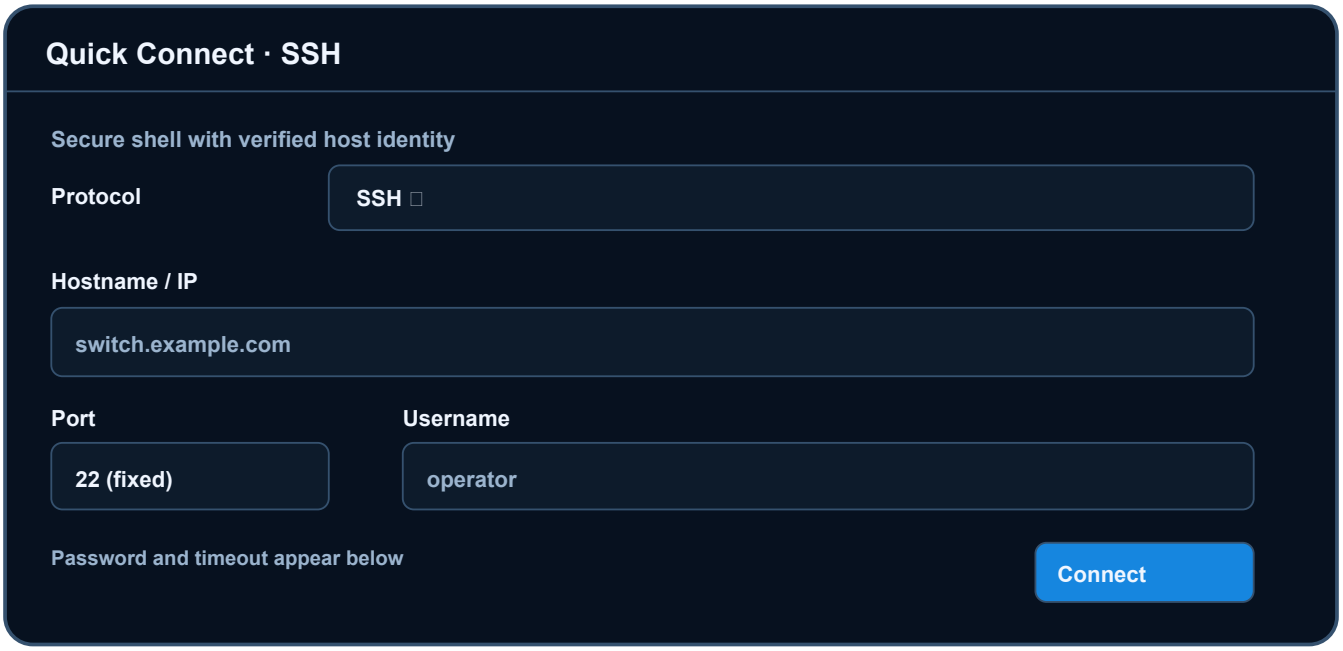


Illustration of the SSH Quick Connect form

On the first SSH connection to a host, inspect the displayed SHA256 host-key fingerprint and confirm it against a trusted source before accepting it. A changed saved key shows a separate warning. Verify the new fingerprint independently before using **Replace saved key**. The app never silently replaces a changed key.

If an older SSH device ignores Backspace, try **Standard**, then **Compatibility: BS (Ctrl+H)**, then **Compatibility: DEL** in the Quick Connect or saved-device connection options. Press the Backspace key normally after changing the setting.

4. Save a device and credentials

Use **Add device** on the left rail for an entry you will reuse. The editor has **General**, **Connection**, **Credentials**, and **Organization** tabs.

1. On **General**, enter a device name. Optionally add description, device type, environment, Favorites status, and notes.
2. On **Connection**, choose **CLI only**, **GUI only**, or **CLI + GUI**. For CLI access, choose a protocol and complete its fields. SSH is fixed to port 22 and Telnet to port 23.
3. For SSH, open **Credentials** and choose a reusable credential profile. Create one from **Credential Profiles** on the rail if needed. You can leave **Username override** blank to use the profile username.
4. On **Organization**, enter a **Site** and, if useful, one **Organization folder** under it. Click **Save device**.

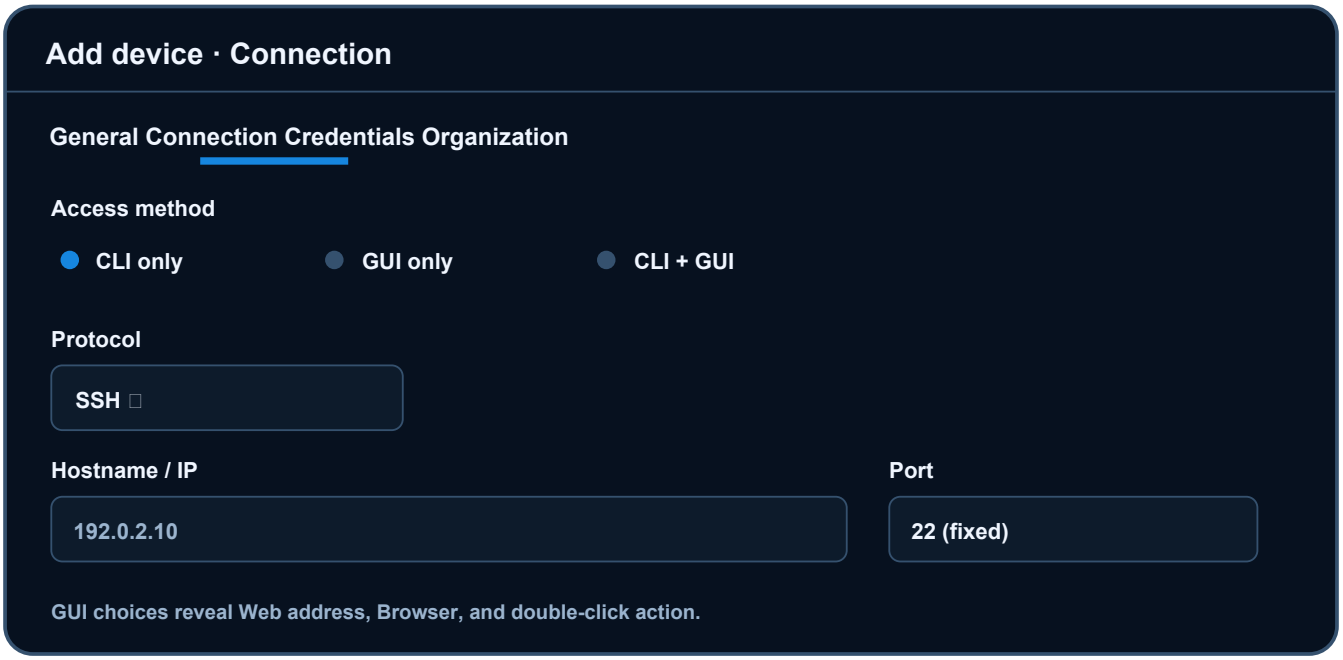


Illustration of the saved-device connection tab

Credential profile passwords are protected by Windows Credential Manager. A saved device refers to its profile; the password is not stored in the device database. **GUI only** does not need an SSH credential. Existing devices can be edited from Device Explorer. Changing a CLI device from SSH to Telnet also requires the Telnet plaintext acknowledgment.

Connection choices

Access method	What appears in the editor	What opening does
CLI only	Protocol and terminal settings	Opens a terminal session
GUI only	Web address and browser	Opens the external browser
CLI + GUI	Both sets of settings	Uses your selected double-click action; both remain in the device menu

5. Open a device's web interface

For **GUI only** or **CLI + GUI**, enter the device's full **http://** or **https://** address in **Web address**. Choose **Windows default browser**, **Microsoft Edge**, or **Google Chrome**. On a device with both methods, set **Open on double-click** to **CLI terminal** or **Web GUI**.

Click **Check access** only if you want to test whether the web port responds. This check does not prove the web page is the right device, validate a TLS certificate, or sign in. Web sign-in stays in your external browser. Prefer HTTPS where the device supports it.

In Device Explorer, double-click a web device or use **Open web interface** from its menu. For a dual-access device, use the menu to open the other method when needed.

6. Organize devices

Device Explorer supports **Site** → **Organization folder** → **devices**. Devices may also sit directly under a site or **Unassigned**. The explorer starts in alphabetical order until you set a custom order.

- Right-click inside Device Explorer and choose **New site**. To create an Organization folder, select a site or a row within it, right-click, and choose **New Organization folder**.
- Drag a device with the left mouse button onto a site or Organization folder to move it. Its saved Site and Organization values update. You can also edit those fields directly on the device's **Organization** tab.
- Hold **Ctrl** or **Shift** to select several device rows, then drag a selected row onto a site or folder to move the selection together. Review the selected rows before dropping.
- Drag a device above or below another device to set a custom order. A blue insertion line shows the destination. Sites and sibling folders can also be reordered.
- To return one site's or folder's direct children to alphabetical order, select that row, right-click, and choose **Sort alphabetically**.
- Use **All**, **Favorites**, **Recent**, and the search field to narrow the list. For moving or ordering rows, return to **All** and clear the search.
- To delete one or several selected devices, select them, use the delete action, and check the count in the confirmation dialog. Deleting a nonempty site or folder is refused until its devices are moved or deleted.

A site is the first level and an Organization folder is the second; the editor does not create deeper folder paths.

7. Use terminal tabs and panes

Each connection opens its own terminal tab, including a tab while the connection is still in progress. This lets you see and cancel an individual attempt. Click a tab to make it active; drag its label left or right to reorder it. The tab and selected-session status dots use green for running and red for disconnected sessions. A connecting tab has its own connecting state.

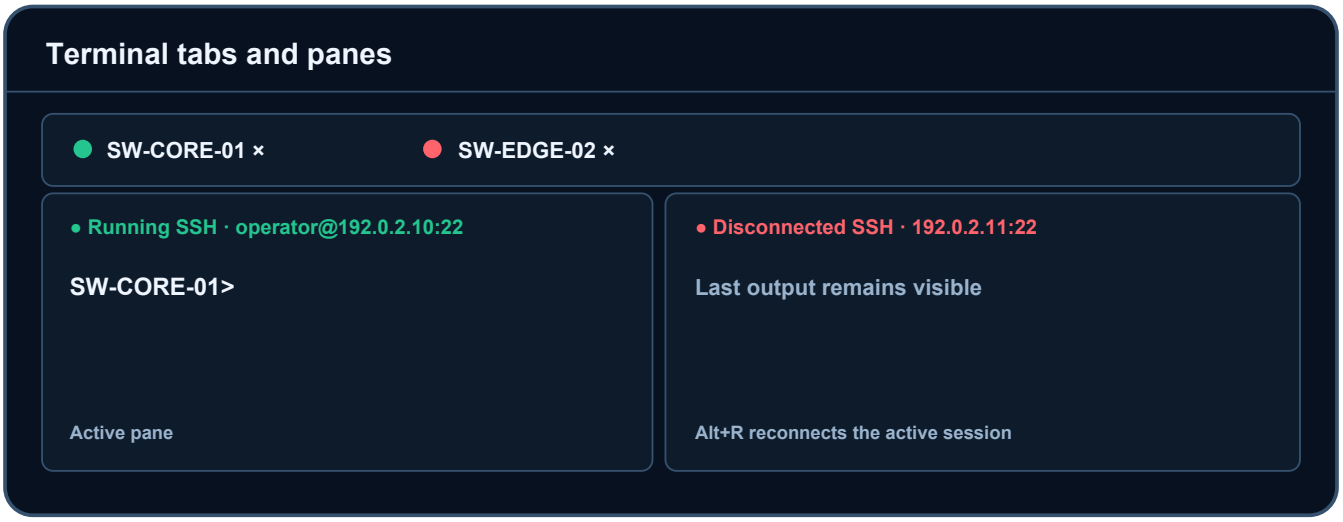


Illustration of terminal tabs, status, and a split pane

Open **PowerShell** or **Command Prompt** from the left rail for a local shell. For the active terminal, use **Alt+Right** to split right or **Alt+Down** to split below. Use **F6** and **Shift+F6** to move focus between panes, and **Ctrl+W** to close the active pane. The tab's **×** closes that tab.

Use **Ctrl+F** to search terminal output. **Alt+L** toggles Input Lock; while locked, the terminal continues to display output but blocks typing and paste. **Alt+G** turns output logging on or off for the active session. **Alt+R** reconnects, and **Alt+X** disconnects. Review the active tab before invoking a session shortcut. A multiline or control-sequence paste opens a confirmation dialog so you can review it first.

8. Import and export devices

Use **Import devices** on the left rail for an Excel workbook, CSV, or JSON device file. A blank Excel import template is available from **Settings**. The template has a **Devices** sheet with dropdowns for accepted values, including Device Type, Protocol, Access Mode, Browser Preference, and Default Open Action, and an **Instructions** sheet.

1. Complete **Name** and **Host** for each row. Use the template's dropdowns for other fields. For Telnet, explicitly set **Protocol = Telnet**, **Port = 23**, and the Telnet acknowledgment. SSH uses port 22 and does not need the Telnet acknowledgment.
2. Import the file and inspect **Review device import** before proceeding. The list shows each device's protocol and endpoint.
3. Choose **Keep both as copies**, **Skip imported duplicates**, or **Update matching devices**. Select a **Fallback SSH credential** if imported SSH devices need one; matching saved credentials may already be used.
4. Click **Import devices** and read the added, updated, skipped, and credential-required counts.

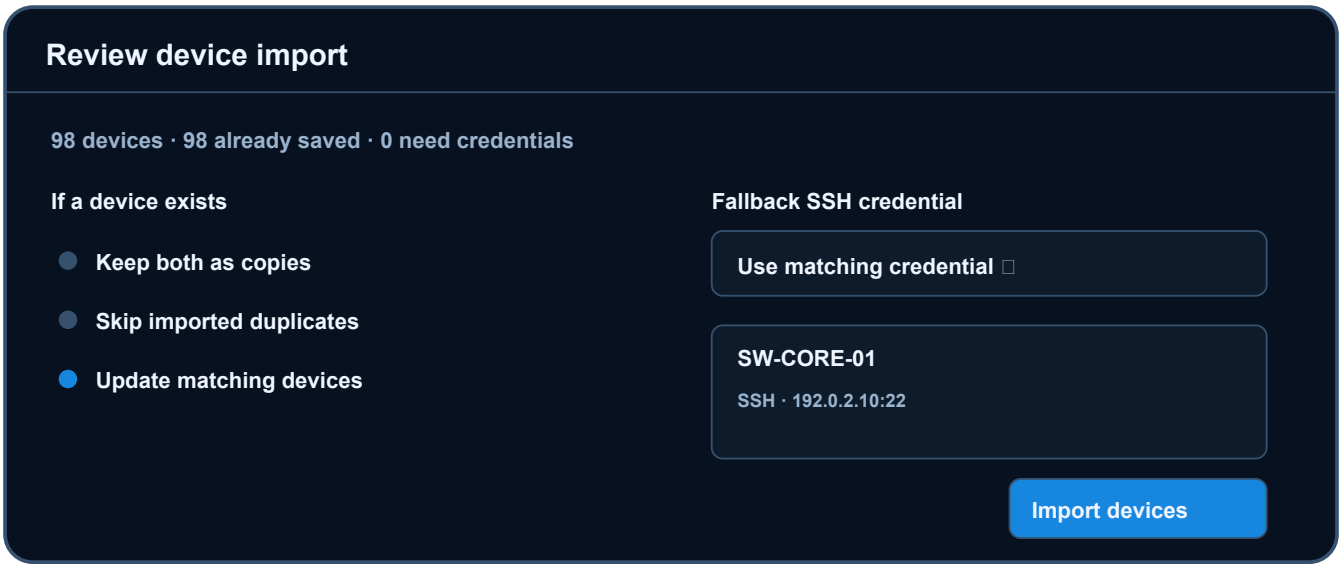


Illustration of the import review choices

For a second import that changes existing device metadata, use **Update matching devices**. A port number alone does not change the protocol: set the Protocol column explicitly. Rows with an SSH port other than 22 or a Telnet port other than 23 are rejected. Imported passwords and trusted host keys are never accepted from a file.

Use **Export devices** to create a JSON device file or Excel workbook. Choose the export scope and whether to include Favorites and Recent metadata. Exported files contain device metadata, including saved web addresses; they exclude passwords, private keys, credential secrets, and trusted host keys. Check exported data before sharing it because names, hosts, sites, notes, and web addresses may identify your infrastructure.

9. Settings and local data

Open **Settings** from the bottom of the left rail to adjust terminal appearance and behavior, find the import template, and open the logs folder. The application uses a permanent dark theme. Settings and device data are local to each Windows account.

Data	Location or owner
Saved devices and workspace metadata	**%LOCALAPPDATA%\NetOpsTerminal\data\netops-terminal.db**
Non-secret preferences	**%LOCALAPPDATA%\NetOpsTerminal\settings.json**
Saved SSH passwords	Windows Credential Manager
Trusted SSH host keys	**%LOCALAPPDATA%\NetOpsTerminal\ssh\known_hosts**
Optional session output logs	**%LOCALAPPDATA%\NetOpsTerminal\session-logs**

Session logging is opt in. Remote devices can echo passwords or other sensitive text, so protect log files and review them before sharing. Workspace restore brings back disconnected tabs; it does not automatically reconnect to devices.

10. Fix common problems

What you see	What to check
SSH host key is unknown	Compare the shown SHA256 fingerprint with a trusted source, then choose whether to trust it.
SSH host key changed	Verify the new fingerprint independently. Only then use **Replace saved key**. If unexpected, stop and ask your network administrator.
SSH handshake fails	Confirm SSH is enabled on the device and its firmware supports compatible algorithms. Older devices may need a vendor firmware update. Review sanitized technical details without sharing credentials.
SSH selected but port shows 23	Edit and save the device; version 1.1.0 fixes SSH to 22 and Telnet to 23. In an import file, set Protocol explicitly.
Backspace does nothing	In connection options, try **Compatibility: BS (Ctrl+H)**, then **Compatibility: DEL**; press Backspace normally.
Imported SSH device needs credentials	Assign an existing credential profile or choose a fallback SSH credential during review.
Web page does not open	Check the saved URL, browser selection, local network access, and device web service. **Check access** tests only whether the port responds.
Desktop shortcut is missing	Check whether the installer checkbox was selected and whether Windows **Show desktop icons** is enabled. The Start menu shortcut is a separate option.
Explorer seems to disappear	Click the Device Explorer rail icon, or pin the explorer to keep it open during connections. Your folder expansion choices remain.

If a connection fails, the **Details** button shows sanitized technical information. Include the error category, protocol, and device model when asking for support, but remove real IPs, usernames, and other sensitive values before sharing a report. Support: **lansangan888@gmail.com**.

11. Keyboard shortcuts

Open the in-app **? Help & Shortcuts** flyout for the authoritative list in your installed build.

Action	Shortcut	Action	Shortcut
Quick Connect	Ctrl+N	Reconnect active session	Alt+R
Start/stop logging	Alt+G	Lock/unlock input	Alt+L
Disconnect	Alt+X	Find output	Ctrl+F
Split right	Alt+Right	Split down	Alt+Down
Close active pane	Ctrl+W	Next pane	F6
Previous pane	Shift+F6	Open PowerShell	Alt+P
Open Command Prompt	Alt+C	Open Help	Ctrl+/
Close Help	Esc		

Version: 1.1.0 · **Publisher:** skadooooooooooosh · **Platform:** Windows 10/11 x64